THE DIVISOR-RESTRICTED GÁL FORM OVER SQUARE-FREE SETS

ØISTEIN SØVIK

This proof note has evidence level Sketch: it has not appeared on arXiv or been peer-reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> « Le résultat suivant, relatif aux éléments de $\mathcal{A}$, laisse supposer que la taille maximale de $Q(M)$ lorsque M décrit les sous-ensemble de $\mathbb{N}_1$ de taille N vaut $\Gamma^*(N)^{1/2+o(1)}$. Cette question demeure ouverte en l'état. Rappelons la définition des normes $Q(M)$ en $(1 \cdot 6)$. »

In §5 the source uses $Q(M)$ for the divisor-restricted quadratic-form norm, which we denote by $\mathcal{Q}(M)$ below. The introduction uses the same notation $Q(M)$ for the full Gál-form norm; we keep that name for the latter in the proof. The quoted closing sentence points back to equation $(1 \cdot 6)$, which defines the full norm, so this is a source-notation discrepancy; the question's scope follows the restricted form defined immediately before the quoted passage. The target ranges over square-free N-element sets, not only $M \in \mathcal{A}$.

Source: de la Bretèche and Tenenbaum, "Sommes de Gál et applications," §5.1, p. 18 (version 8, 2022; published in Proceedings of the London Mathematical Society 119 (2019), doi:10.1112/plms.12224, arXiv:1804.01629v8). The inferred question concerns the maximum of the divisor-restricted form over square-free N-element sets.

Here $\mathbb{N}_1$ denotes the square-free integers, and $\mathcal{A}$ consists of finite square-free sets M whose prime factors are at most the $|M|$th prime. For a finite set $M \subset \mathbb{N}$, define

$$Q(M) = \sup_{\|c\|_2=1} \Big| \sum_{m,n\in M} c_m c_n \frac{(m,n)}{\sqrt{mn}} \Big|,$$

$$\mathcal{Q}(M) = \sup_{\|c\|_2=1} \Big| \sum_{\substack{m,n\in M \\ n\mid m}} c_m c_n \sqrt{n/m} \Big|,$$

$$\Gamma^*(N) = \sup_{\substack{|M|=N \\ M \text{ square-free}}} \frac{1}{N} \sum_{m,n\in M} \frac{(m,n)}{\sqrt{mn}},$$

$$\mathscr{L}(x) = \exp \sqrt{\frac{\log x \log_3 x}{\log_2 x}}.$$

Here (m,n) is the greatest common divisor and $\log_j$ is the j-fold iterated logarithm; x is sufficiently large for the displayed expression.

Answered: yes. Evidence: Sketch.

Yes. Over square-free sets the maximum is $\Gamma^*(N)^{1/2+o(1)}$.

Theorem 1. *For every finite $M \subset \mathbb{N}$ with $|M| = N \geq 2$,*

$$\mathcal{Q}(M) \leq 1 + \sqrt{E(N^2)Q(M)}, \qquad E(x) = \prod_{p\leq x} (1 - 1/p)^{-1}.$$

Corollary 1. *As $N \to \infty$, the maximum of $\mathcal{Q}(M)$ over square-free N-element sets is $\Gamma^*(N)^{1/2+o(1)}$.*

Date: 13 October 2026.

Proof of Theorem 1. For coefficient vectors a, b indexed by M, write

$$\Sigma = \sum_{\substack{m,n\in M\\ n\mid m}} a_m b_n \sqrt{n/m}.$$

Put $y = N^2$. Split the pairs (m,n) with $n \mid m$ according to whether m/n is y-smooth (every prime factor at most y) or not, and write the sum as $\Sigma_1 + \Sigma_2$ accordingly.

The rough part. If m/n has a prime factor larger than y, then $m/n > N^2$ and $\sqrt{n/m} < 1/N$. The diagonal pairs are smooth, so each row and each column of the matrix of Σ_2 has at most $N-1$ nonzero entries, each less than $1/N$. The Schur test gives $|\Sigma_2| \le \|a\|_2 \|b\|_2$.

The smooth part. Let P be the set of primes dividing some element of M, let $P_y = \{p \in P : p \le y\}$, and write $\beta_p(n)$ for the exponent of p in n. On $\mathbb{T}^P$ with Haar measure σ set

$$F(z) = \sum_{n\in M} b_n \prod_{p\in P} z_p^{\beta_p(n)}, \qquad G(z) = \sum_{m\in M} \overline{a_m} \prod_{p\in P} z_p^{\beta_p(m)},$$

$$K(z) = \prod_{p\in P_y} \frac{1}{1 - p^{-1/2} z_p}.$$

Expanding K in geometric series, the coefficient of $\prod_p z_p^{\beta_p(m)-\beta_p(n)}$ in K equals $\sqrt{n/m}$ when $n \mid m$ and m/n is y-smooth, and 0 otherwise. Indeed, it vanishes unless $\beta_p(m) \ge \beta_p(n)$ for $p \in P_y$ and $\beta_p(m) = \beta_p(n)$ for $p \in P \setminus P_y$. Hence

$$\Sigma_1 = \int_{\mathbb{T}^P} FK\,\overline{G}\,\mathrm{d}\sigma, \qquad |\Sigma_1| \le \|G\|_{L^2} \|FK\|_{L^2} = \|a\|_2 \, \|FK\|_{L^2}.$$

For $r = p^{-1/2}$ the Poisson kernel $P_p(z) = (1-r^2)|1 - rz|^{-2}$ has Fourier coefficients $r^{|k|}$, by expanding two geometric series. Since $|1 - p^{-1/2} z_p|^{-2} = (1 - 1/p)^{-1} P_p(z_p)$,

$$\|FK\|_{L^2}^2 = \prod_{p\in P_y} \Big(1 - \frac{1}{p}\Big)^{-1} \sum_{n,n'\in M} b_n \overline{b_{n'}}\, \kappa(n,n'),$$

where $\kappa(n,n') = 0$ unless $\beta_p(n) = \beta_p(n')$ for all $p \in P \setminus P_y$, in which case

$$\kappa(n,n') = \prod_{p\in P_y} p^{-|\beta_p(n) - \beta_p(n')|/2}.$$

The Gál kernel $(n,n')/\sqrt{nn'}$ is the same product taken over all $p \in P$. Hence $0 \le \kappa(n,n') \le (n,n')/\sqrt{nn'}$ in both cases, and

$$\Big|\sum_{n,n'} b_n \overline{b_{n'}} \kappa(n,n')\Big| \le \sum_{n,n'\in M} |b_n||b_{n'}| \frac{(n,n')}{\sqrt{nn'}} \le Q(M) \|b\|_2^2,$$

the last step by taking $c = |b|/\|b\|_2$ in the definition of $Q(M)$. As P_y consists of primes at most y, the product is at most $E(N^2)$, and $|\Sigma_1| \le \sqrt{E(N^2) Q(M)}\, \|a\|_2 \|b\|_2$. Adding the two parts proves the inequality. Taking $a = b = c$ with $\|c\|_2 = 1$ gives the bound for $\mathcal{Q}(M)$. □

Proof of Corollary 1. By Mertens' theorem $E(N^2) \sim 2e^\gamma \log N = \mathscr{L}(N)^{o(1)}$. Theorem 1.2 of de la Bretèche and Tenenbaum, "Sommes de Gál et applications," doi:10.1112/plms.12224, gives $\sup_{|M|=N} Q(M) = \mathscr{L}(N)^{2\sqrt{2}+o(1)}$, and Theorem 1 then gives the first bound. For square-free sets, equation (1.7) of de la Bretèche and Tenenbaum and the sentence following it give $\sup Q(M) \le (e^2+1)(\log N + 2) \max_{n\le N} \Gamma^*(n)$ over N-element $M \subset \mathbb{N}_1$. By equation (1.5) of de la Bretèche and Tenenbaum the right side is $\mathscr{L}(N)^{2+o(1)}$, so Theorem 1 gives $\mathcal{Q}(M) \le \mathscr{L}(N)^{1+o(1)}$ for these M. The matching lower bound is the lower bound in equation (5.5) of de la Bretèche and Tenenbaum, attained by square-free sets; the source credits the construction in Bondarenko and Seip, "Large GCD Sums and Extreme Values of the Riemann Zeta Function," arXiv:1507.05840, §2, Lemmas 1–2 and the divisor-closed construction. Finally $\mathscr{L}(N)^{1+o(1)} = \Gamma^*(N)^{1/2+o(1)}$ by equation (1.5) of de la Bretèche and Tenenbaum. □